

Tribhuvan University
Faculty of Management
M. Phil. in Public Administration, 2077

First Semester Examination
MPH 502: Research Methods in Public Administration-II

Full Marks: 100
Time: 4 hours

Read the instruction and answer accordingly

This exam paper is divided into two parts. Part A is theory-based and Part B computer based.

Unless and otherwise stated, all notations/ terminologies have their usual meanings

Answer sheet of both part must be submitted

Part “A”

[Full Marks: 30]

[time: two hours]

Attempt any five questions

1. The Ministry of Federal Affairs and General Administration wishes to know the average income of general assistants. A sample of 60 assistants shows a sample mean of Rs17,400 with a standard deviation of Rs. 3,150. (a) Place a 90% confidence limit around your best estimate of the average income of general assistance. (b) What is the probability that the average income could be Rs18,500 or more? (c) What is the probability that the average income might be as low as Rs.17,000?
2. The Siddhartha Art Gallery wants to know whether its parking situation is getting worse. Specifically, administrators are concerned that a lack of parking is driving away potential visitors. The administrators plan to address the situation if at least 15% of visitors feel that finding parking near the facility is a problem (the center averages about 3,100 visitors per week). To get a sense of whether this is really the case, the administrators distribute a brief exit survey to 150 individuals who visit the center over the following week. The results indicate that 32 of 150 visitors complained about being unable to find adequate parking. Present alternate and null hypothesis. Calculate a z score and use it to evaluate the hypotheses. What should the administrators conclude from these results?
3. Two scholars are locked in debate regarding the interpretation of the accompanying data. One insists that the relationship between the age of a child and the child's perception of the parents is very strong; the other argues that no relationship exists. A survey was conducted on 200 children to know their perception of parents and its output is presented in the table given below. Analyze the table and resolve the dilemma: Is there a relationship between the age of a child and the child's perception of the parents?

Child's Perception of Parents	Age of Child (Years)		
	5–15	16–27	28 and Older
Negative	22	106	46
Neutral	36	54	118
Positive	142	40	36
Total	200	200	200

4. What do you mean by Multivariate data analysis? Shed light on exploratory factor analysis.
5. Differentiate between parametric and non-parametric test of Hypothesis. What are the advantages and disadvantages of non-parametric tests?
6. A research on the differences between men and women on overall attitude level says the difference is “statistically significant.” Discuss the meaning of statistical significance. A research on attitude score gives the following result after performing in SPSS.

Group Statistics

Attitude score of	N	Mean	Std Deviation	Standard error of mean
Male	39	55.79	9.339	1.495
Female	59	59.49	7.835	1.020

Independent Sample Test

	Levene's Test for Equality of Variance		Test for Equality of Means		
	F	Sig	t	df	Sig (2 tail)
Equal variance assumed	4.317	0.040	2.117	96	0.037
Equal variances not assumed			2.042	71.5	0.045

- i) What statistical tool was used in this test?
 - ii) Write down all the steps needed to come to the decision about this problem.
 - iii) What changes do you find in the decision if the significance level was changed from 5% to 1%?
7. The Federal Government of Nepal wishes to measure the significance difference on satisfaction level of employees in the different ministries. For this purpose three ministries: Ministry of Youth and sports, Ministry of Science and Technology and Ministry of Water supply were selected and their satisfaction level was measured using a tested tool. The following data shows the satisfaction level:
- Ministry of Youth and Sports: 23, 22, 59, 26, 27, 66, 51, 39
- Ministry of Science and Technology: 38, 50, 29, 49, 58, 46, 56, 37
- Ministry of Water Supply: 48, 60, 30, 52, 55, 44, 65, 62
- Using Kruskal-Wallis H test at 0.05 level of significance, test the hypothesis that average satisfaction level of employees of different ministries are different?

Table of critical values

Degree of freedom	0.2	0.1	0.05
2	3.219	4.605	5.991
3	4.642	6.251	7.815
4	5.989	7.779	9.488

[Full Marks: 30]

[time: two hours]

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All questions are compulsory and each carries 15 marks

8. Recently a research was conducted to study the citizen report card of selected VDC's of Chitwan and Nawalparasi district of Nepal. Some portion of this data is given to you with the file name CRC2020.sav. On the basis of this data, Operate the different function in SPSS and answer the followings:

Complete tables showing

- i) the number of respondent, mean and standard deviation of monthly income by their education level

Monthly Income			
Education	N	Mean	Standard Deviation
Below SLC/SEE			
SLC/SEE			
Inter/+2			
Bachelor			
Masters and above			
Total			

- i) the number of respondents, mean and standard deviation of monthly income by the district where they reside.

Monthly Income			
Sex	N	Mean	Standard Deviation
Male			
Female			

- III) Have you find any significant difference in the average monthly income among different education levels? (Use appropriate test of hypothesis)

Value of test statistics	P-value	Conclusion

national product (GNP). Two such indicators being examined are government deficit (in Lakhs Rs.) and industrial average. Data for 8 year are given below which is also given with the file name NPC.sav.

Percentage change in GNP (Y)	2.5	1	4	1	1.5	3	5	4.5
Government Deficit (x1)	80	60	90	65	70	90	112	108
Industrial average (x2)	950	700	1100	720	850	900	1200	1150

On the basis of this data, Operate the different function in SPSS and answer the followings:

i) Obtain the unstandardized regression coefficients of percentage change in GDP on Government Deficit and Industrial average along with standard error, t-value and significant and fill the table given below.

	Unstandardized Coefficients		t	Sig
	Coefficients (bi)	Standard error (sbi)		
Constant				
Government Deficit (x1)				
Industrial average (x2)				

ii) Interpret the values of the regression coefficients so obtained and also discuss the significance of these values

Regression Coefficients	Interpretation / Significance

iii) Obtain the fitting regression equation for these data and find the percentage change in GNP would be expected in a year with government deficit of Rs 120 Lakhs and industrial average of 1000.

Regression Equation:
Percentage change in GNP=? When government deficit = Rs 120 Lakhs and industrial average =1000

iv) Obtain R- Square value and Standard error of the model. What percentage of variation in GNP is explained by this equation?

R- square	
Standard Error	
What percentage of variation in GNP is explained by this equation?	

v) Fill the ANOVA table of the model so obtained and discuss the overall model fit.

Source	df	SS	MS	F	Sig
Regression					
Residual					
Total					
Discussion on the model					